ML-Driven Inverse Design of Microring Resonators for Co-Packaged Optics / High-Bandwidth Optical Interconnects

Md Shidul Islam Shihab*[1]

[1]Department of Electronics & Communication Engineering, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

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Abstract

Deep-learning inverse design of photonic components rests on an unresolved disagreement. Liu *et al.* [3] showed that a naive, non-tandem inverse network can fail to converge when the design space is redundant; Chen and Jiang [7] and Adibnia *et al.* [9] reported naive inverse networks converging acceptably in their own design spaces. This work resolves the disagreement for a four-parameter, bus-coupled add-drop silicon microring resonator by measuring both sides of it on one dataset with one set of metrics. We build a dual-fidelity dataset of 1,150 labelled geometries over ring radius, waveguide width, coupling gap and coupling length, with the coupling coefficient obtained from 200 three-dimensional FDTD simulations of the coupler alone and the ring response computed analytically, and we validate the construction against a published, experimentally characterised device, against 68 further independent FDTD runs, and against nine full 3-D simulations of the complete ring.

The naive inverse network *fails* the geometry-space criterion ($R^2 \geq 0.80$ on only one of four parameters) and *passes* the response-space criterion (closed-loop re-simulated error 3.669%, all twenty test designs physically realisable). Both prior reports are therefore correct, and were measuring different things. We explain why by counting constraints: of four target quantities, resonance wavelength carries almost no geometric information ($R^2 = 0.0078 \pm 0.0011$ across five seeds), free spectral range fixes the round-trip length, and loaded Q and insertion loss both act through a single coupling phase θ — two effective constraints for four unknowns, leaving a two-dimensional degenerate family.

Three further results follow. First, geometry-space error does not predict response-space error: across four models and three architecture families, closed-loop error is proportional to θ error to within a factor 1.20, while the same ratio formed with RMS-RGE varies by a factor 4.00. Second, physical realisability is architecture-dependent and is invisible to both metrics — Random Forest emits buildable geometries in 100% of cases, provably so by convexity, against 95.4% for a plain network and 33.5% for a tandem network. Third, a reproducible information bound computed from a 10^6-geometry response bank places the achievable RMS-RGE at or below 14.626%, which establishes that the 15% objective threshold lies above the information content of four scalar targets and that the measured shortfall to 15.506% is a model-and-data result rather than a fundamental limit. Translating the designs into two co-packaged-optics scenarios, no method meets any scenario's full specification; eleven of twelve failures involve resonance placement, whose closed-loop error of 4.32 nm RMS exceeds a ±2.0 nm channel tolerance by a factor of two

*mdshidulislamshihab@gmail.com

and a ± 0.4 nm tolerance by eleven. Absolute resonance placement in this design space is beyond inverse design and requires post-fabrication trimming.

1 Introduction

Inverse design asks for a geometry given a desired optical response. Where a forward simulation is expensive, a natural approach is to train a surrogate on simulated data and then invert it, and a substantial literature now does exactly this for nanophotonic components [1,2]. Within that literature sits a specific, unresolved disagreement about whether the simplest approach works.

Liu *et al.* [3] argued that training a network to map response directly to geometry fails when several geometries produce the same response: the network is asked to fit a one-to-many relation and converges to an average that is itself not a solution. Their remedy, the tandem network, trains the inverse network through a frozen forward model so that the loss is measured in response space, where the relation is single-valued. This argument is widely accepted and widely cited.

Against it, two later studies report the naive approach working. Chen and Jiang [7] inverse-designed a four-ring channel-dropping filter with two independently trained networks — no tandem — and reported good convergence, validating several designs by re-simulation. Adibnia *et al.* [9] likewise reported acceptable convergence for a direct inverse network. Neither is a refutation of [3], and [3] is not a refutation of them, because the three studies work in different design spaces and, critically, judge success by different criteria: [3] reports geometric accuracy, while [7] and [9] report how closely a re-simulated design reproduces the requested response.

No prior study resolves this for a microring resonator. Pal *et al.* [6] is the only work applying classical tree-based machine learning to microring inverse design, on 462 finite-element samples, but their design space contains no bus-coupling parameter at all and so cannot predict insertion loss or any other coupling-dependent quantity — the specifications that decide whether a filter is usable in a co-packaged-optics (CPO) link. Torabi *et al.* [12], the most recent and most dimensionally comparable study, defines the RMS-RGE metric adopted here but addresses a polymer-scale biosensor ring and includes no classical-machine-learning baseline; it is also an unrefereed preprint and is cited as such throughout. Valencia-Garzón *et al.* [10] release a large all-pass dataset without modelling it.

This paper contributes the following.

1. **A resolution of the disagreement**, measured on one dataset with one protocol: the naive inverse network fails in geometry space and succeeds in response space, so both camps are right. We give a closed-form reason, by counting how many independent constraints four scalar targets actually impose.
2. **Evidence that geometry-space error is the wrong metric**, and a physically motivated replacement that predicts response-space error across architectures.
3. **A measurement of physical realisability as a first-class result**, with a proof that one model family cannot violate the design box.
4. **A reproducible information bound** on inverse-design accuracy for this design space, which separates model limitations from information limits.
5. **A controlled four-way comparison** of Random Forest, XGBoost, a plain deep network and a tandem network on an identical split with identical metrics, including a sample-efficiency characterisation — the comparison absent from [4,6,7,9,12].

Table 1: Design space and target quantities.

Symbol	Parameter	Range	Note
R	ring radius	5 µm to 15 µm	spans FSR ≈ 5.7 nm to 18.2 nm
w	waveguide width	400 nm to 500 nm	single-mode strip on 220 nm SOI
g	coupling gap	150 nm to 350 nm	near-critical to weakly coupled
L_c	coupling length	0 µm to 3 µm	$L_c = 0$ is a valid design point
λ_{res}	resonance wavelength	nm	nearest to 1550 nm
FSR	free spectral range	nm	adjacent-resonance spacing
Q_L	loaded quality factor	—	λ_{res}/FWHM
IL	insertion loss	dB	drop port, on resonance

2 Device, design space and methods

2.1 Device and parameterisation

The device is a two-bus, four-port add-drop racetrack microring resonator on silicon-on-insulator: silicon core of fixed height 220 nm, SiO_2 cladding, fundamental TE mode only, scanned over 1500 nm to 1600 nm with a nominal target of 1550 nm. Both couplers share a gap and coupling length, keeping the design space at exactly four dimensions. Dispersive material models are used throughout; no constant index is assumed. A racetrack rather than a circular ring makes the coupling length a genuine continuous parameter, which is the degree of freedom absent from [6], [10] and [12].

Table 1 lists the four free parameters and the four targets. The round-trip length is $L = 2\pi R + 2L_c$.

2.2 Dual-fidelity dataset construction

A full three-dimensional FDTD simulation of the complete ring was measured at roughly 35 h per sample, which puts a 1,000-sample dataset beyond any realistic budget. A two-dimensional simulation runs in 149 min but returns a loaded Q too high by a factor of 11.3, because it cannot represent vertical confinement.

The construction used instead simulates in three dimensions only the part of the device that genuinely requires it. A single pass of the directional coupler is simulated in 3-D FDTD to obtain the power coupling coefficient κ^2; the ring response is then assembled analytically from coupled-mode theory [13,14], with the effective and group indices taken from a 3-D mode sweep and the propagation losses from a bend-loss sweep. The total cost is approximately 16 h.

A five-parameter model is fitted to the coupler simulations,

$$\kappa^2 = \sin^2\theta, \qquad \theta = \kappa_0(w)\, e^{-\gamma(w) g} \left(L_c + B\sqrt{2\pi R/\gamma} \right), \tag{1}$$

in which the gap and the coupling length enter *only* through the single phase θ. Equation (1) is the origin of the degeneracy analysed in Section 3.2, and θ is the quantity used as a metric in Section 3.3. The fitted $B = 1.0211$ is close to its first-principles value of unity.

The coupler set was built in two stages by augmented Latin hypercube sampling: 48 points, then a further 152, for 200 in total. The extension widened the κ^2 range from 0.0021–0.3138 to 0.0019–0.4599 and moved B from 1.0425 toward unity, with residuals flat across two and a half decades and no evidence of $\sin^2\theta$ saturation.

The modelling dataset comprises 1,150 labelled geometries — 1,000 from a primary Latin hypercube design plus a 150-point supplementary batch covering a sparse corner of the space —

Table 2: Validation of the dual-fidelity construction. The out-of-sample κ^2 tests are the coupler simulations performed during closed-loop re-verification (Section 3.3); they were not available to the fit.

Test	Quantity	Result	Criterion
Published device [11]	FSR, 440 nm×220 nm, $R = 20\,\mu$m	4.4008 nm vs 4.400 nm measured	$\leq 10\%$ (O1)
	implied n_g	4.3443 vs 4.3451	—
κ^2 model	8-fold cross-validated MAPE	3.36%, $R^2 = 0.9976$	—
κ^2 model	68 out-of-sample FDTD runs	MAPE 1.23–5.16%, all within 10%	—
	systematic bias	-0.36 to $+3.33\%$	—
Energy conservation	$\kappa^2 + t^2$	1.0000 on every run	—

split 805/172/173 into training, validation and test sets with a fixed seed. The test set was used once.

2.3 Validation of the construction

Table 2 summarises three independent checks. The predicted free spectral range of the fabricated device reported by Tang *et al.* [11] agrees with the measured value to within the precision of the published figure (quoted as 4.4 nm, i.e. $\pm 1.1\%$), well inside the 10% requirement. Two caveats must accompany this validation. Only FSR is tested, and FSR depends on group index and round-trip length alone, so the test exercises the mode solver and the length convention and says nothing about the coupling model, the bend-loss model, Q_L or IL. Further, the propagation loss used in the analytic model (1.3 dB/cm) is itself taken from [11]; it does not enter the FSR prediction, so this particular comparison is unaffected, but [11] is not an independent source for the loss model. We also note that the loaded $Q \approx 3 \times 10^5$ appearing in [11] is a stated assumption in a scalability projection, not a measurement, and is therefore not used here.

The κ^2 model was further tested on 68 coupler geometries that arose during closed-loop re-verification and were never seen by the fit. Every one fell within 10%, with a consistent positive bias of a few per cent that should be reported as a known systematic.

2.4 Full 3-D validation of the ring model

The checks above validate the *coupler* and the *mode solver*. The coupled-mode assembly that turns κ^2, n_g and loss into Q_L and IL is not tested by either, and closed-loop re-verification does not test it because it re-simulates the coupler and then recomputes the ring with the same analytic model that produced the labels. Since Q_L and IL carry the degeneracy argument of Section 3.2, we simulated the *complete ring* in 3-D FDTD.

Nine runs were performed: three geometries at $R = 5.20\,\mu$m, $w = 450$ nm, spanning the coupling range, each at the production 32 nm mesh, at a coarse 40 nm mesh, and at 40 nm with three times denser spectral sampling as a control on the extraction. The simulation window was sized from the predicted Q_L so that the linewidth exceeds the transform limit of the DFT window by at least a factor three, and auto-shutoff was disabled so that the window is exactly the one assumed. All nine met the resolution criteria (6.1–52.9 points across the FWHM, 3.14–4.12 times the transform limit).

Three findings follow, and they are not all favourable.

Resonance wavelength and free spectral range behave as the published-device check predicted. λ_{res} agrees to 0.10% across all nine runs. FSR is biased +2.2 to +2.8%, the same sign and magnitude every time. Since $\mathrm{FSR} = \lambda^2/(n_g L)$ and $L = 2\pi R + 2L_c$ is exact geometry, the real ring's group index is about 2.5% below the model's. The model takes n_g from a *straight-waveguide*

Table 3: Full 3-D ring simulation against the analytic model. Richardson extrapolation assumes second-order convergence, which two mesh sizes cannot verify.

	simulated Q_L				analytic	
Geometry	32 nm	40 nm	6000 pts	$h \to 0$	Q_L	error
$g = 150$ nm, $L_c = 2.00$ µm	1780	1749	1751	**1833**	1855	**−1.2%**
$g = 170$ nm, $L_c = 1.00$ µm	4015	3892	3921	**4182**	4264	**−1.9%**
$g = 180$ nm, $L_c = 0.60$ µm	4603	5229	5253	3447	6351	−45.7%

mode sweep, and a 5.2 µm bend is not a straight waveguide; a bent-waveguide mode solve would correct this but would change every dataset label, so it is left to future work.

The validation itself was mesh-limited, and at low to moderate Q_L the analytic model is correct to within 2%. Simulated Q_L depends on the mesh, so the 32 nm spacing — established as converged for κ^2, a single-pass quantity — is not converged for Q_L, which accumulates loss over hundreds of round trips. For the two lower-Q_L geometries, refinement moves the simulation *toward* the analytic prediction and a second-order extrapolation to zero mesh recovers it to 1.2% and 1.9% (Table 3). A discrepancy of 4–6% at the production mesh is therefore numerical, not physical.

At the highest Q_L the discrepancy is real and unexplained. For the third geometry, refinement moves the simulation *away* from the model. The tripled spectral sampling changed Q_L by 0.45%, so this is neither an extraction nor a sampling artifact. One candidate is that the bend-loss model is a function of radius alone and carries no wavelength dependence, while this geometry resonates at 1555 nm against 1542 nm and 1543 nm for the other two; bend radiation at this radius increases steeply toward longer wavelengths, so a wavelength-blind loss model would over-predict Q_L for the reddest resonance, and a finer mesh resolving genuine radiation would lower it. We state this as a hypothesis: testing it requires a wavelength-resolved bend-loss study that we have not performed.

Insertion loss does not validate. Drop-port peak transmission differs from the analytic prediction by −12.5%, +1.5% and −17.9% at the production mesh, with no consistent sign. One of the three is not evidence at all: that geometry's IL is 0.084 dB, i.e. $T = 0.981$, within 2% of unity, where the absolute uncertainty of extracting IL from a simulated spectrum is comparable to the value itself.

What this does and does not affect. Every model compared in Section 3 was trained and scored on identical labels, so a bias shared by all of them cancels from every model-to-model comparison: the resolution of O6, the metric result of Section 3.3, the realisability result and the sample-efficiency comparison are unaffected. What must be qualified is the absolute accuracy of Q_L above $Q_L \approx 4300$, the absolute accuracy of IL anywhere, and the scenario verdicts, which are judged against Q_L specifications.

2.5 Models and protocol

Four method families are compared, each in the forward and inverse direction, on the identical split with identical metrics: Random Forest, XGBoost, a plain feed-forward network, and a tandem network trained through the frozen forward model. Tree models are evaluated with raw and normalised inputs. The forward network architecture was selected by a 144-configuration grid search over depth, width, learning rate and weight decay.

Two metric conventions require statement. Aggregate geometric error is the root-mean-square

Table 4: Objective outcomes. O6 is the central contribution; O1–O5 and O7 exist to make its resolution credible rather than to stand as independent results of equal weight.

	Outcome	Basis
O1 dataset + validation	met	1,150 geometries ($\geq$ 800 required); FSR agrees with [11] to within the published figure's precision, well inside 10%
O2 forward surrogate	**not met**	2 of 4 targets; IL MAPE 11.331 ± 1.101% against ≤ 10% over five seeds. Cause: small denominators on a sub-decibel target, at 0.072 dB RMSE
O3 inverse accuracy	**not met**	RMS-RGE = 15.506% against $\leq$ 15%; closed-loop half met at 3.669%. Reachable in principle: information bound ≤ 14.626%
O4 four-way comparison	met	all four families, identical split, every metric, no omissions
O5 sample efficiency	met	XGBoost forward below 300 samples; network at full size; naive network inverse at both
O6 one-to-many	**resolved**	geometry-space criterion fails (1 of 4 parameters at $R^2 \geq 0.80$), response-space criterion passes (3.669%) — both prior reports correct
O7 CPO translation	**not met**	0 of 12 candidates, FDTD-verified; 11 of 12 failures involve resonance placement, which is physically out of reach

relative geometric error

$$\text{RMS-RGE} = 100\% \times \sqrt{\frac{1}{4} \sum_{p \in \{R, w, g, L_c\}} \left(\frac{p^{\text{pred}} - p^{\text{target}}}{s_p} \right)^2}, \tag{2}$$

with $s_p = p^{\text{target}}$ for R, w and g following [12], and $s_p = 3\,\mu\text{m}$, the range width, for L_c — necessary because $L_c = 0$ is a valid design point at which a target-normalised relative error is undefined. Per-parameter terms and the pure [12]-convention value over $\{R, w, g\}$ are reported alongside every aggregate.

Closed-loop response error is measured by taking a model's predicted geometry for each of twenty representative test targets, re-simulating the coupler in 3-D FDTD, reassembling the ring response, and comparing to the response originally requested. The FDTD step is what makes this independent: feeding predictions back through the fitted κ^2 model would merely replay the model that produced the labels.

3 Results

Table 4 summarises the outcome against the seven objectives fixed before any modelling began. Three are met, three are not, and the central one is resolved in the direction that the methodology document anticipated as publishable. The objectives that are not met are reported as measured, with the reason in each case identified rather than asserted.

3.1 The forward surrogate, and an unlearnable target

The best forward model is the network, at 3.568% mean MAPE across the four targets against 5.323% for the best XGBoost variant and 11.929% for the best Random Forest. Table 5 reports all

Table 5: Forward-direction accuracy on the held-out test set ($n = 173$). Every method appears for every target. The best network is reported as a mean over five training seeds, since the objective threshold is a hard bound and a single run cannot distinguish a crossing from initialisation noise.

	R^2				MAPE (%)			
Method	λ_{res}	FSR	Q_L	IL	λ_{res}	FSR	Q_L	IL
DNN	0.0069	0.9993	0.9980	0.9969	0.158	0.561	3.205	10.346
RF (raw)	−0.0950	0.6892	0.9504	0.9342	0.166	17.288	12.640	25.351
RF (norm)	−0.0688	0.9920	0.9147	0.9072	0.164	2.730	17.340	27.483
XGB (raw)	−0.0837	0.9992	0.9634	0.9494	0.168	0.699	8.811	12.640
XGB (norm)	−0.0795	0.9992	0.9808	0.9489	0.168	0.695	7.701	12.727
DNN over five seeds (42–46), mean ± s.d.:								
λ_{res}	$R^2 = 0.0078 \pm 0.0011$				MAPE $= 0.158 \pm 0.000$			(0/5 seeds pass)
FSR	$R^2 = 0.9993 \pm 0.0001$				MAPE $= 0.556 \pm 0.020$			(5/5 pass)
Q_L	$R^2 = 0.9982 \pm 0.0001$				MAPE $= 3.059 \pm 0.109$			(5/5 pass)
IL	$R^2 = 0.9966 \pm 0.0003$				MAPE $= 11.331 \pm 1.101$			(1/5 pass)

three required metrics for every target and method.

Objective O2 required $R^2 \geq 0.90$ and MAPE $\leq 10\%$ for at least three of four targets. **Two of four are met, so O2 is not met.** Two points about that verdict deserve care.

First, the verdict is reproducible, and it was measured as such. λ_{res} fails on R^2 by an enormous margin and does so consistently; IL fails on MAPE at $11.331 \pm 1.101\%$ against a 10% bar, with only one seed in five passing. A single training run is not a sound basis for a hard-threshold verdict, and we report the distribution rather than a draw from it.

Second, IL's failure is a property of the metric and the target's scale, not of the model's accuracy. IL is predicted to 0.072 dB RMSE with $R^2 = 0.9966$. But the dataset's IL has a median of 0.524 dB, with 48.3% of rows below 0.5 dB and a minimum of 0.032 dB, so a constant absolute error of 0.07 dB produces 13.7% relative error at the median and over 200% at the extreme. Clearing a 10% MAPE bar would require absolute accuracy better than about 0.052 dB on a decibel quantity already predicted to seven hundredths of a decibel. We report O2 as not met under the criterion as defined, and note that 0.072 dB sits inside typical CPO link-budget tolerance.

The resonance wavelength is not learnable by any method here. All four tree variants return *negative* R^2 — worse than predicting the mean — and the network manages 0.0069. The cause is sensitivity: $\mathrm{d}\lambda_{\text{res}}/\mathrm{d}R \approx 0.15\,\text{nm/nm}$, so placing a resonance to within 1 nm requires the radius to roughly 7 nm, about one part in 1.5×10^4 of the range. The consequence propagates: λ_{res} accounts for almost all of the forward model's residual loss, so the 144-configuration grid search was in effect ranking configurations on noise.

3.2 The one-to-many question, resolved

Objective O6 requires the naive inverse network to be judged against *both* criteria used by the two sides of the disagreement: a geometry-space criterion of $R^2 \geq 0.80$, and a response-space criterion of closed-loop re-simulated error.

In geometry space the naive network fails. Its per-parameter R^2 values are 0.9930 for R, 0.2920 for w, 0.6914 for g and 0.2166 for L_c: one parameter of four clears 0.80. Figure 1 shows that this is not specific to the network — every method recovers the radius and no method recovers the other three.

In response space the naive network passes, comfortably. Re-simulating its predicted

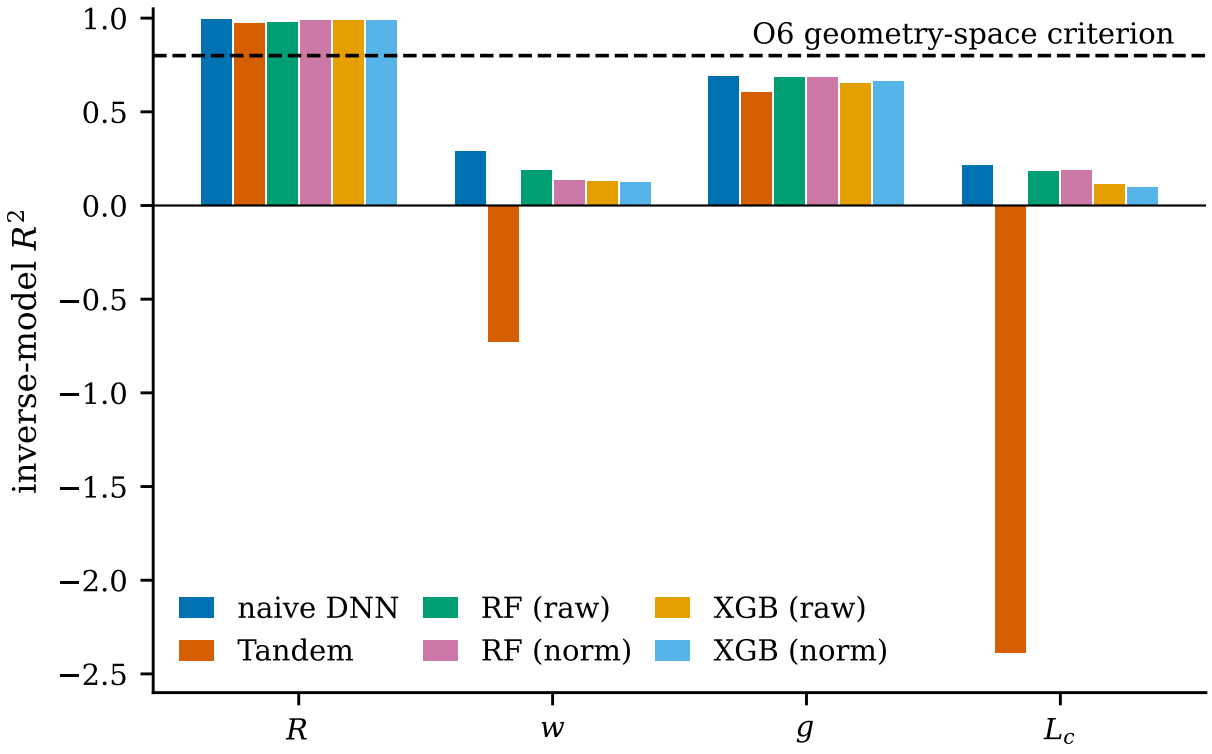


Figure 1: The geometry-space half of the one-to-many test. Only the ring radius is recovered; width, gap and coupling length are not, for every method. The dashed line is the $R^2 \geq 0.80$ criterion. The tandem network's negative values for w and L_c indicate predictions worse than the test-set mean.

geometries for twenty representative targets gives a root-mean-square closed-loop response error of 3.669%, against a 20% requirement, with all twenty predictions physically realisable. Per target: λ_{res} 0.203%, FSR 0.980%, Q_L 4.037%, IL 3.594%.

This is the outcome in which the two literatures are simultaneously correct. Liu *et al.* [3] are right that the naive network does not recover the geometry; Chen and Jiang [7] and Adibnia *et al.* [9] are right that it reproduces the requested response. The disagreement was never empirical.

Why: counting constraints. The structure of the problem explains the split exactly. Of the four targets, λ_{res} carries essentially no geometric information, as Section 3.1 established. FSR fixes the round-trip length $L = 2\pi R + 2L_c$. And Q_L and IL both depend on the coupling only through $\kappa^2 = \sin^2\theta$, so they constrain the single phase θ of Eq. (1) and nothing finer. Four targets therefore impose two effective constraints — one on L, one on θ — on four unknowns, leaving a two-dimensional family of geometries with indistinguishable responses.

That prediction is quantitatively borne out. L is dominated by $2\pi R$, so R is recoverable, and it is ($R^2 = 0.9930$). The remaining three parameters enter only through θ, and none is individually recoverable ($R^2 = 0.13$–0.69 across all methods). Direct confirmation: four geometries with gaps spanning 210 nm to 270 nm and coupling lengths spanning 0.209 µm to 2.353 µm share a loaded Q of 14 659.8 to the last digit. The naive network recovers θ itself to 1.824%, with 96.5% of its predictions within 5% of the correct level set. It is not failing; it is landing on the right level set and being scored on its position along it.

Table 6: Inverse-direction accuracy, all methods on the identical test split. RMS-RGE is Eq. (2); per-parameter terms and the $\{R, w, g\}$-only value are given as the metric convention requires. "Buildable" is the fraction of test predictions lying inside the design box.

	RMS-RGE (%)						
Method	aggregate	R	w	g	L_c	R, w, g only	Buildable
naive DNN	**15.506**	2.841	5.321	14.144	26.930	8.878	95.4%
Tandem	29.461	6.079	7.852	15.296	56.029	10.529	33.5%
RF (raw)	16.039	5.942	5.741	14.378	27.460	9.574	100%
RF (norm)	15.835	3.730	5.926	14.213	27.421	9.148	100%
XGB (raw)	16.451	2.949	5.925	14.847	28.607	9.385	100%
XGB (norm)	16.546	2.986	5.941	14.687	28.900	9.308	100%

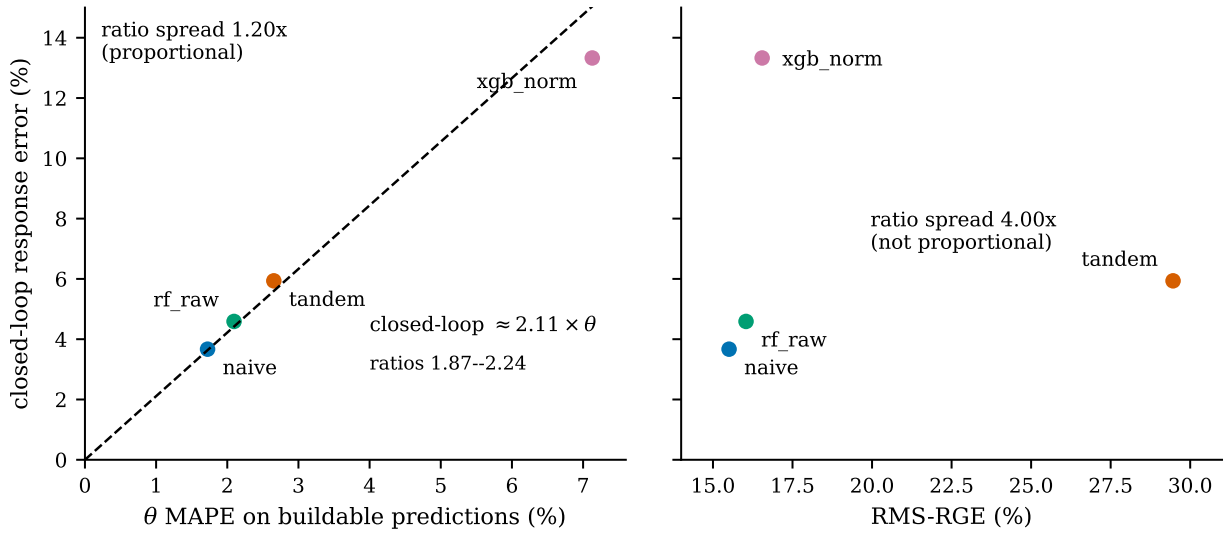


Figure 2: Closed-loop response error against two candidate metrics, for the four models whose predictions were re-simulated. Left: error in the coupling phase θ, evaluated on buildable predictions so that the two axes cover the same designs. Right: RMS-RGE. The ratio of closed-loop error to θ error is constant to within a factor 1.20; formed with RMS-RGE it varies by a factor 4.00.

3.3 Geometry-space error does not predict response-space error

Section 3.2 showed that geometric error and response error disagree about whether a model works. Table 7 shows that they also disagree about which models work, and by how much.

The naive network and XGBoost (normalised) differ by 1.04 percentage points in RMS-RGE — 15.506% against 16.546%, a difference a reader would reasonably call negligible. In closed loop they differ by a factor of 3.6: 3.669% against 13.329%. Had only RMS-RGE been measured, these two models would have been called equivalent, and the conclusion would have been wrong by nearly a factor of four on the quantity that determines whether a design works.

The degeneracy suggests why, and θ makes it measurable. An error that moves along a θ level set yields a different valid design; one that moves across it yields a wrong response. Figure 2 tests this on the four models whose predictions were re-simulated. Closed-loop error is proportional to θ error, with the four ratios spanning 1.87–2.24, a factor of 1.20. The same ratios formed with RMS-RGE are 0.20–0.81, a factor of 4.00.

Table 7: Closed-loop re-verification, four models, twenty targets each. All clear the 20% response-error bar. θ error is evaluated on buildable predictions, matching the designs over which closed-loop error is measured.

Method	RMS-RGE (%)	θ err. (%)	closed-loop error, per target (%)				RMS (%)
			λ_{res}	FSR	Q_L	IL	
naive DNN	15.506	1.723	0.203	0.980	4.037	3.594	**3.669**
RF (raw)	16.039	2.095	0.208	2.994	4.058	5.011	4.589
Tandem[†]	29.461	2.654	0.190	0.745	4.980	7.110	5.936
XGB (norm)	16.546	7.129	0.165	0.970	14.684	15.382	13.329

[†]Over the 8 of 20 predictions that were buildable; see Section 3.4.

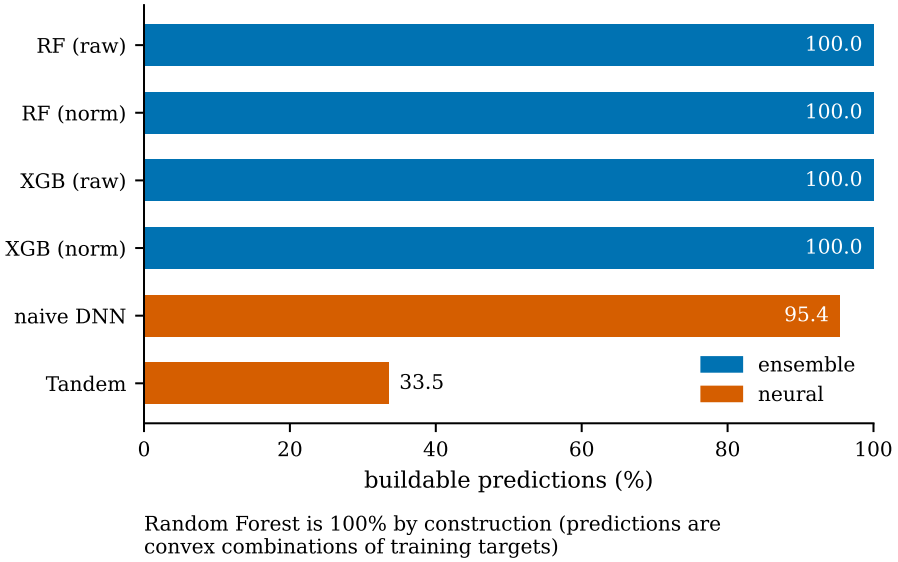


Figure 3: Fraction of test-set predictions lying inside the design box. The distinction is architectural, and invisible to both metrics of Section 3.3.

Two features of that comparison matter for its strength. The first is that model family is controlled: Random Forest (raw) is an ensemble method with θ error 2.095% and lands at 4.589%, beside the network's 3.669% and far from XGBoost's 13.329%. Within the tree family alone, θ error varies by a factor 3.4 and closed-loop error by 2.9, while RMS-RGE differs by 3% — so there is no architecture explanation available. The second is that ordering alone would not have discriminated: RMS-RGE happens to rank three of these four models correctly. What it cannot do is say how much worse one is than another, which is what a design metric is for.

3.4 Physical realisability

Neither RMS-RGE nor θ error detects a prediction that cannot be fabricated, and the differences between architectures here are large.

The tandem network produced unbuildable geometries for twelve of the twenty closed-loop targets: eleven with negative coupling length, down to $-0.881\,\mu\text{m}$, and one with a width of 397.8 nm. Over the full test set its buildable fraction is 33.5%. The mechanism is visible in its loss: the tandem network is penalised only through the frozen forward surrogate, which will evaluate $L_c = -0.88\,\mu\text{m}$ and return a plausible response. Nothing in the objective encodes that the design space has edges.

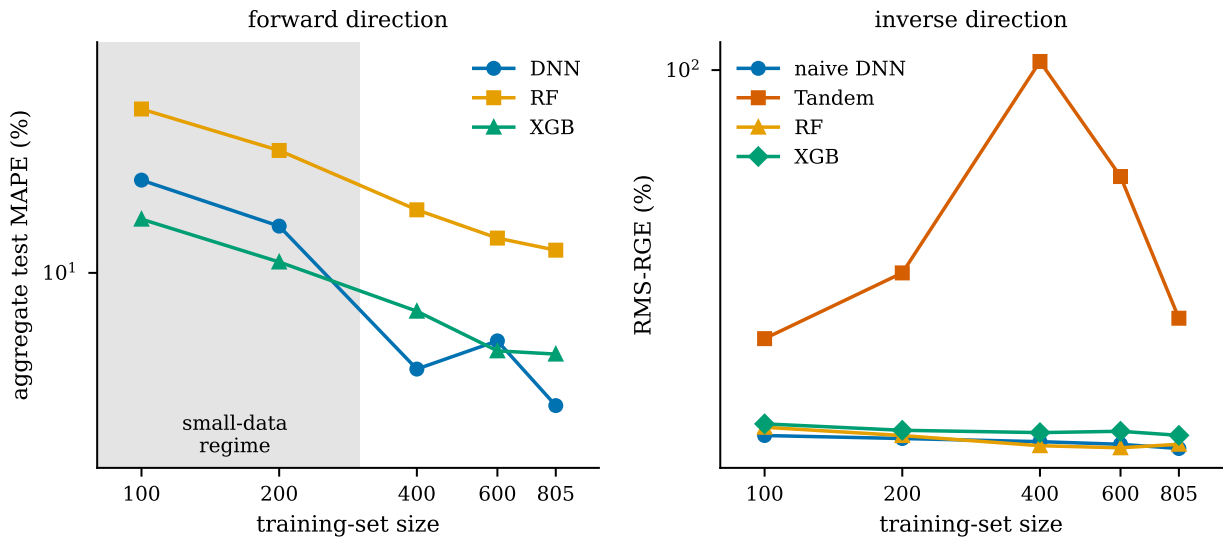


Figure 4: Sample efficiency. Left: in the forward direction XGBoost is more accurate below about 300 training samples and the network overtakes it by the full dataset. Right: in the inverse direction the naive network is best at every size, and the tandem network is unstable, peaking at 104.3% at 400 samples.

The naive network, which regresses against real geometries and is therefore pulled toward a region where all targets are valid, leaves the box on 4.6% of the test set.

Both tree families produced buildable geometries in every case. For Random Forest this is guaranteed rather than fortunate: its prediction is a mean of leaf values and hence a convex combination of training targets, so it lies in their convex hull, and the design box is convex. Gradient boosting is additive rather than convex and carries no such guarantee, but violated the box nowhere here.

The practical consequence is that θ error and realisable fraction are complementary and both are needed; neither subsumes the other. The tandem network illustrates why. On its buildable predictions its closed-loop error is 5.936%, better than XGBoost's, and its θ error is a respectable 2.654%. Judged on those numbers alone it looks like a competitive model. Sixty per cent of its designs cannot be made.

3.5 Method comparison and sample efficiency

In the forward direction, XGBoost is the most accurate method in the ≤ 300-sample regime, at 13.030% mean aggregate MAPE over sizes 100 and 200 against the network's 17.465%; at the full 805-sample training pool the network is most accurate, at 3.568%. In the inverse direction the naive network is lowest at both ends — 16.411% mean RMS-RGE in the ≤ 300-sample regime and 15.506% at full size.

The crossover is worth stating plainly because it bears on practice: for a photonic design problem where each label costs hours of simulation, a gradient-boosted tree is the better forward surrogate until a few hundred samples have been gathered. Pal *et al.* [6] drew their result from 462 samples and Torabi *et al.* [12] from 2,500, so this range spans the operating point of the closest published work.

The tandem network's inverse curve is not a learning curve at all: 26.6%, 36.8%, 104.3%, 59.2%, 29.5% across the five sizes. This is consistent with the box-escape mechanism of Section 3.4; with

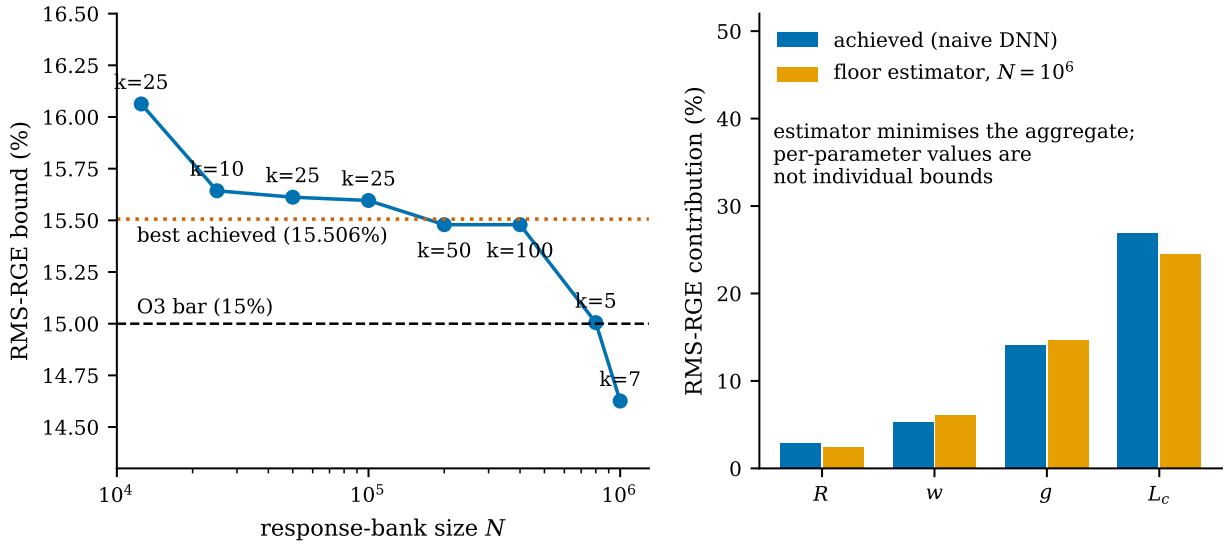


Figure 5: Left: the information bound against response-bank size, with the best neighbourhood size k annotated. The estimate is an upper bound on the true bound, and falls as the bank grows. At $N = 10^6$ it lies below the 15% objective threshold. Right: per-parameter error of the best model against the bound estimator's per-parameter error; these are not individual bounds, since the estimator minimises the aggregate.

nothing anchoring predictions inside the design space, where the network wanders is largely a function of initialisation and data.

A final observation on metric dependence. Ranking parameter difficulty by mean R^2 gives $L_c > w > g > R$, hardest first; ranking by mean RMS-RGE contribution gives $L_c > g > w > R$. The two orderings disagree on w against g. Under either, the coupling gap is *not* the hardest parameter — the coupling length is, with mean $R^2 = -0.2645$ — which refutes the hypothesis that g would dominate by analogy with [6]'s finding for radius.

3.6 An information bound

Objective O3 required, for at least one method, RMS-RGE $\leq 15\%$ *and* closed-loop error $\leq 20\%$. The closed-loop half is met with a wide margin (3.669%). The geometric half is not: the best aggregate RMS-RGE is 15.506%. **O3 is therefore not met**, by 0.506 percentage points.

Whether that shortfall reflects the models or the problem is a separate question, and answering it requires knowing how much geometric information four scalar targets contain. We measure this directly. A bank of 10^6 geometries is drawn over the design box and labelled with the same analytic responder that produced the dataset, subject to the same quality gate. For each test target, the k bank geometries whose responses are closest are averaged, and the RMS-RGE of that average against the true geometry is computed. Because this is the error of one specific predictor, it is an *upper bound* on the Bayes-optimal error.

The asymmetry of that bound governs what can be concluded. An upper bound measured below a threshold proves the true bound is below it; measured above, it proves nothing. Figure 5 shows the estimate falling from 16.063% at $N = 12\,500$ to 14.626% at $N = 10^6$. Since 14.626% < 15%:

> **O3's geometric threshold lies above the information content of the four chosen targets' inverse problem. The objective is reachable in principle, and**

Table 8: Re-simulated scenario verdicts. Each candidate geometry was re-simulated with a fresh 3-D FDTD coupler run. Tolerance on λ_{res} is half the channel spacing: ± 2.0 nm in A, ± 0.4 nm in B.

Method	λ_{res} (nm)	FSR (nm)	Q_L	IL (dB)	Verdict
Scenario A — 4 channels, 4 nm *grid; FSR* ≥ 16 nm, $Q_L \in [1500, 8000]$, *IL* ≤ 1.5 dB					
naive DNN	1546.892	16.351	3995	0.352	fail: λ_{res} −3.11 nm
Tandem	1543.385	16.640	3826	0.345	fail: λ_{res} −6.61 nm
RF (raw)	1548.058	15.869	5425	0.462	fail: FSR −0.13 nm
RF (norm)	1543.823	15.901	5917	0.508	fail: λ_{res} −6.18, FSR −0.10
XGB (raw)	1557.673	16.298	3708	0.322	fail: λ_{res} +7.67 nm
XGB (norm)	1547.746	16.111	3890	0.336	fail: λ_{res} −2.25 nm
Scenario B — 8 channels, 0.8 nm *ITU grid; FSR* ≥ 6.4 nm, $Q_L \geq 7750$, *IL* ≤ 1.0 dB					
naive DNN	1548.663	6.458	8575	0.207	fail: λ_{res} −1.34 nm
Tandem	1548.770	6.344	8196	0.190	fail: λ_{res} −1.23, FSR −0.06
RF (raw)	1549.537	8.446	10571	0.361	fail: λ_{res} −0.46 nm
RF (norm)	1548.733	6.474	12291	0.301	fail: λ_{res} −1.27 nm
XGB (raw)	1550.934	6.405	6070	0.147	fail: λ_{res} +0.93, Q_L −1680
XGB (norm)	1552.793	6.424	5816	0.141	fail: λ_{res} +2.79, Q_L −1934

the measured shortfall is a model-and-data result, not a fundamental limit. At least 0.879 percentage points of headroom separate the best model from the bound.

Two honest qualifications. The estimate is not converged: the optimal neighbourhood size drifts from $k = 25$ to $k = 7$ across the sweep, and the descent accelerates from 0.096 to 0.530 percentage points per e-fold of bank size. A shifting optimum means the estimator's own finite-sample bias is still unwinding, so no rate from this sweep describes the true bound, and a three-parameter asymptotic fit to these points is not identifiable. And the per-parameter values in Figure 5 are not individual bounds: the estimator minimises the aggregate and trades parameters against one another, which is why the best model legitimately beats two of them.

We also record a negative decision. Enlarging the training set with additional rows drawn from the *fitted* analytic model lowers measured RMS-RGE below 15%. Those rows add sampling density, not physics — the dataset would still carry the information of 200 coupler simulations — and the closed-loop error against FDTD would not improve. We therefore do not report O3 as met by that route, and record here that it was available and declined.

3.7 Translation to co-packaged-optics scenarios

Objective O7 asked which methods meet a CPO scenario's specification within its stated tolerance. Table 8 answers: **none, in either scenario**. The verdicts rest on fresh FDTD re-simulation of each candidate rather than on the analytic model.

Eleven of the twelve failures involve resonance placement. The single exception is Random Forest (raw) in Scenario A, which placed the resonance at 1548.058 nm, inside the ± 2.0 nm window, and then missed FSR by 0.13 nm. FSR, Q_L and IL are otherwise largely satisfied; only XGBoost undershoots Q_L in Scenario B.

This failure is physical, not algorithmic, and the numbers say so without interpretation. The naive network's closed-loop λ_{res} error is 3.15 nm MAPE, 4.32 nm RMS and 10.9 nm at worst. Scenario A allows 2.0 nm and Scenario B 0.4 nm, so the best model misses by factors of two and eleven respectively. Working back through $\mathrm{d}\lambda_{\text{res}}/\mathrm{d}R \approx 0.15$ nm/nm, hitting Scenario A would require

the radius controlled to about 13 nm and Scenario B to about 3 nm — below realistic lithographic tolerance, let alone inverse-model precision.

Three independent measurements agree on this: the sensitivity analysis of Section 3.1, the closed-loop re-simulation of Section 3.3, and the scenario translation here. Inverse design delivers the *shape* of the response — free spectral range, loaded Q, insertion loss — and cannot deliver absolute resonance position. A CPO deployment must place channels by post-fabrication thermal or carrier trimming.

4 Discussion

The central result is that a question treated as empirical in the inverse-design literature was in fact definitional. Asking whether a naive inverse network "converges acceptably" has no answer until one says in which space the error is measured, and the two available answers differ here by a factor of four on the same predictions. For a degenerate design space the geometry-space question is not merely harder, it is the wrong question: a model that lands on the correct level set has produced a valid design, and penalising its position along that set measures nothing a user cares about.

That observation generalises beyond microrings. Wherever the forward map compresses several geometric degrees of freedom into fewer physical ones — which is to say, wherever inverse design is interesting — the same split should be expected, and the recommendation follows directly: report closed-loop response error, and report it alongside a physically meaningful invariant of the degeneracy rather than a raw geometric distance. Our θ is such an invariant for this device; the general construction is to identify the quantity through which the degenerate parameters act and measure error in it.

The realisability result adds a second axis that neither error metric sees. It also puts a specific cost on the tandem architecture. The tandem network was introduced [3] precisely to handle degeneracy, and in response space it does: its closed-loop error on buildable designs, 5.936%, is better than XGBoost's. But measuring the loss only through a surrogate removes the sole pressure keeping predictions inside the manufacturable domain, and two thirds of its designs leave it. A tandem network for a bounded design space needs an explicit constraint, and reporting realisable fraction would have made this visible in earlier work.

The information bound changes how a missed threshold should be read. RMS-RGE targets in this literature are quoted without reference to how much geometric information the chosen targets contain — [12] reports 3.46% and 5.14% under the $\{R, w, g\}$ convention, against 8.878% here under the same convention, in a design space one to two orders of magnitude different in three of four dimensions. Such numbers are not commensurable, and a bound computed per design space is what makes them interpretable. Here the bound establishes that the shortfall is addressable; in another design space the same computation might show a target to be unreachable, which is equally useful to know before pursuing it.

5 Limitations

1. **The ring validation is not mesh-converged, and cannot be on the available hardware.** Simulated Q_L depends on mesh spacing (Section 2.4). A third, finer mesh for these geometries exceeds the 15.7 GB of system memory available, so the Richardson extrapolation rests on two mesh sizes and assumes second-order convergence, which two points cannot verify.

2. **Ring validation is at small radius only.** All nine runs are at $R = 5.20\,\mu\text{m}$. A 10 µm ring requires about 20 GB of solver memory at this mesh and a 14 µm ring about 36 GB, so the radius

range could not be spanned. Small radius is where bend loss is largest and the loss model most stressed, but the validation does not establish accuracy at large radius.

3. Q_L **accuracy is established only below** $Q_L \approx 4300$, to within 2%. Above it the model over-predicts by 17–27% and the cause is not established. **IL is not validated at all.**

4. **The external validation covers FSR only.** It tests the mode solver and the length convention. The coupling model, the bend-loss model, Q_L and IL are not externally validated against measurement.

5. **The information bound is not converged.** It is a valid upper bound at every bank size, which is enough for the one-directional conclusion drawn, but its asymptote is not determined by the sweep performed.

6. λ_{res} **is unlearnable within this parameterisation.** An alternative target — for example resonance offset from a grid wavelength, or azimuthal mode order — might be learnable. We measured one reparameterisation candidate and found that within an already matched set the mean $|\text{corr}(\lambda_{\text{res}}, g)|$ is 0.095, so λ_{res} is close to redundant with FSR for geometry recovery; a systematic study was not performed.

7. **Both couplers share a gap and coupling length.** Asymmetric couplers are a valid extension and would change the degeneracy structure.

8. **Closed-loop verification uses twenty targets per model**, the stated minimum, and four of six models. The two unmeasured models are ones for which the two candidate metrics make predictions differing by less than one percentage point.

9. **No fabrication.** Every result is simulation-based. The resonance-placement conclusion in particular predicts that trimming is required; it does not demonstrate a trimmed device.

10. **No independent expert review of the physical methods** was obtained prior to this draft.

6 Conclusion

For a four-parameter add-drop silicon microring resonator, a naive inverse neural network fails the geometry-space convergence criterion and passes the response-space one. Both findings reported in the prior literature are therefore correct, and the apparent disagreement between them was a difference of metric rather than of result. The reason is a two-dimensional degeneracy that follows from counting constraints: four scalar targets impose two effective conditions, one on round-trip length and one on a single coupling phase.

Three consequences for practice follow. Geometric error does not predict response error — two models within 1.04 percentage points of each other in RMS-RGE differ by a factor of 3.6 in closed loop, while error in the coupling phase predicts closed-loop error to within a factor 1.20 across three architecture families. Physical realisability is a separate and architecture-dependent property that neither metric detects, ranging from a provable 100% for Random Forest to 33.5% for a tandem network. And an information bound computed per design space is what makes an accuracy target interpretable; here it shows a missed 15% threshold to be addressable rather than fundamental.

For co-packaged optics specifically, the useful conclusion is a negative one established three independent ways: inverse design can deliver free spectral range, loaded Q and insertion loss, and cannot place an absolute resonance wavelength to channel-grid accuracy. Channel assignment must

be achieved by post-fabrication trimming, and an inverse-design flow for CPO filters should be specified on that basis.

Data and code availability

The complete simulation, training and evaluation pipeline is a single Python script, and every reported number is produced by one of its documented modes. The dataset, the fitted models, the response bank and all report files will be released with the manuscript.

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